

CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL COIMBATORE - INDIA

ASSESSMENT OBJECTIVES FOR ENGLISH FOR CLASS XI(CBSE)

- I. COMPREHENSION, ANALYSIS, INFERENCE
II. CREATIVE AND ANALYTICAL WRITING SKILLS
III. ADEQUATE VOCABULARY AND GRAMMAR**

I. COMPREHENSION, ANALYSIS, INFERENCE

There will be a comprehension passage or a poem and exercises will be given based on it. They need not be purely comprehension questions, but may test the students' ability to draw inferences.

II. CREATIVE AND ANALYTICAL WRITING SKILLS

The candidate appearing for the exam should be able to write a fairly lengthy composition/ story on a given topic.

The student is also expected to know both formal and informal letter writing.

An article to a press

III. ADEQUATE VOCABULARY AND GRAMMAR

Transformation of sentences (simple, compound, complex sentences)

Direct / Indirect speech, Active + Passive voice

Vocabulary : a) Analogies b) Words in context

The entrance examination will test the student in the above areas. These areas are broadly delineated cannot be specified in a detailed or precise manner in the curriculum content.

SCHEME OF ASSESSMENT FOR ENGLISH FOR CLASS XI(CBSE)

Assessment Objectives	Weightage (%)	Duration of Paper	Marks
Exercises based on Comprehension	40	1 Hour	50 Marks
Creative/ Critical Writing Skills	30		
Vocabulary Exercises	30		

CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL COIMBATORE – INDIA

ASSESSMENT OBJECTIVES FOR MATHEMATICS FOR CLASS XI

- I. Mathematical knowledge with understanding of concepts
- II. Applications of concepts – Problem solving skill

I. MATHEMATICAL KNOWLEDGE WITH UNDERSTANDING OF CONCEPTS

The candidate taking the entrance exam should be able to present mathematical knowledge and understanding of concepts in relation to:

- a. Organise and present information accurately in written, tabular, graphical and diagrammatic forms.
- b. Perform calculations by suitable methods.
- c. Understand systems of measurement in everyday use and make use of them in the solution of problems.
- d. Estimate, approximate and work to degrees of accuracy appropriate to the context.

II. APPLICATIONS OF CONCEPTS – PROBLEM SOLVING SKILL

The candidate taking the entrance exam should be able to solve the numerical problems using appropriate formulae, symbols, units, graphs etc. The candidate will be tested on the following skills:

- a. Interpret, transform and make appropriate use of mathematical statements expressed in words or symbols.
- b. Recognise and use spatial relationships in two and three dimensions, particularly in solving problems.
- c. Recall, apply and interpret mathematical knowledge in the context of everyday situations.
- d. Make logical deductions from given mathematical data.
- e. Recognise patterns and structures in a variety of situations, and form generalizations.
- f. Respond to a problem relating to a relatively unstructured situation by translating it into an appropriately structured form.
- g. Analyse a problem, select a suitable strategy and apply an appropriate technique to obtain its solution.
- h. Apply combinations of mathematical skills and techniques in problem solving.
- i. Set out mathematical work, including the solution of problems, in a logical and clear form using appropriate symbols and terminology.

The entrance examination will test the above objectives. These skills cannot be further specified in a detailed or precise manner in the curriculum content. However the questions are well within the syllabus of the entrance examination.

SCHEME OF ASSESSMENT FOR MATHEMATICS FOR CLASS XI

Assessment Objectives	Weightage (%)	Duration of Paper	Marks
Mathematical Knowledge	10	1 Hour	50 Marks
Understanding of Concepts	30		
Application Skills	40		
Problem solving skills	20		

SYLLABUS OUTLINE OF MATHEMATICS FOR CLASS XI

Algebra

- The student will
 - a) simplify algebraic expressions using basic operations
 - b) simplify/factorize the algebraic expressions using standard identities $(a + b)^2$, $(a - b)^2$, $a^2 - b^2$, $(a + b + c)^2$, $(a + b)^3$, $(a - b)^3$, $a^3 + b^3$ and $a^3 - b^3$.
 - c) find the quotient and the remainder when a polynomial is divided by another.
 - d) use remainder theorem to find the remainder and factor theorem to factorize a given polynomial of degree at most 3.
 - e) find LCM and HCF of given polynomials.
 - f) determine the roots of the given quadratic equation by factorization / formula.
 - g) frame and solve quadratic equations and linear equations in two variables

Ratio and Proportion and commercial mathematics

- The student will
 - a) solve problems using ratio, proportions and their properties.
 - b) solve word problems using percentage, profit & loss, discount and compound interest.

Measurement and Geometry

- The student will
 - a) find the surface area and volume of cylinder, cone, sphere, hemisphere and frustum of cone.

- b) solve problems involving area of parallelogram, trapezium, rhombus and quadrilateral.
- c) solve simple problems using the properties of quadrilaterals, circles and similarity of triangles.

Statistics

- The student will
 - a) analyze, display, and interpret data in a variety of graphical methods, including line and bar and circle (**pie**) graphs and draw inferences from graphs.
 - b) compute mean, median and mode of discrete data.
 - c) compute mean from a frequency distribution table.

Trigonometry

- The student will
 - a) determine the values of trigonometric ratios from a given right triangle.
 - b) compute the values of trigonometric ratios for special angles such as 30° , 45° , 60° & 90° .
 - c) solve problems using heights and distances with angle of elevation and depression (problems with one or two triangles only)

ASSESSMENT OBJECTIVES FOR PHYSICS FOR CLASS XI(CBSE)

I. Factual knowledge with understanding of concepts

II. Application of concepts – Problem solving

**I. FACTUAL KNOWLEDGE WITH UNDERSTANDING OF CONCEPTS AND
DRAWING SKILLS**

The candidate taking the entrance exam should be able to present factual knowledge and understanding of concepts in relation to:

- Facts, phenomena and Laws.
- Define terms with SI units and its symbols and equations.
- Explain a concept using examples from daily life or laboratory apparatuses (includes drawing of graphs, drawing of apparatus or experimental setups to substantiate a concept).
- Explain or suggest applications of science and technology in daily life with appropriate examples.

II. APPLICATION OF CONCEPTS – PROBLEM SOLVING

The candidate taking the entrance exam should be able to solve the numerical problems using appropriate formulas, symbols, units, graphs etc. The candidate will be tested on the following skills:

- Present the information given in the problem accurately.
- Convert units from one form to another.
- Manipulate data and equations.
- Present the solution with appropriate steps.
- Present the result as desired by the problem with appropriate units.

The entrance examination will test the above objectives. These skills cannot be further specified in a detailed or precise manner in the curriculum content. However the questions are well within the syllabus of the entrance examination.

SCHEME OF ASSESSMENT FOR PHYSICS FOR CLASS XI(CBSE)

Paper	Assessment Objective	Weightage %	Duration of the paper	Marks
Entrance to class 11	Factual Knowledge	10	30 Minutes	25 Marks
	Understanding of Concepts	30		
	Application Skills	40		
	Drawing Skills	20		

NOTE: Separate Papers for Physics, Chemistry and Biology.

SYLLABUS OUTLINE OF PHYSICS FOR CLASS XI(CBSE)

TOPICS: MECHANICS; LIGHT; ELECTRICITY AND MAGNETISM; WAVES AND SOUND

MECHANICS

- Concept of distance, displacement, speed, velocity and acceleration.
- Graphical representation of the above quantities.
- Equations of motion
- Newton's laws of motion.
- Gravitation: Basic concepts.
- Energy – Mechanical Energy; Kinetic Energy; Potential Energy; Conservation of Energy.
- Problems based on the above.

LIGHT

- Basic properties of light.
- Laws of Reflection and Refraction.
- Lenses and Mirrors – Image formation. (Ray Diagrams)
- Problems based on lens and mirror formulae

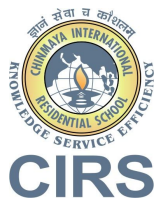
ELECTRICITY AND MAGNETISM

- Concept of electric current, potential difference, resistance – their interrelationship.
- Ohm's Law.
- Resistance in series and parallel (formula for series and parallel combination, circuit diagrams and numerical problems).
- Heating effects of current (formula and problems).
- Conductors and Insulators (qualitative aspects only)

WAVE MOTION AND SOUND

- Transverse and Longitudinal waves.
- Graphical representation of waves.
- Wavelength, amplitude, frequency, time period and speed of the waves.
- Reflection of sound: Applications including ultrasound and sonar.

Questions will also be asked to test the general awareness in science pertaining to the age group.



CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL COIMBATORE - INDIA

ASSESSMENT OBJECTIVES FOR CHEMISTRY FOR CLASS XI(CBSE)

The assessment objectives in Chemistry are:

- A. Knowledge with understanding**
- B. Handling information and solving problems**

A description of each assessment objective is as follows.

A. KNOWLEDGE WITH UNDERSTANDING

Students should be able to demonstrate knowledge and understanding in relation to

1. scientific phenomena, facts, laws, definitions, concepts and theories;
2. scientific vocabulary, terminology and conventions (including symbols, quantities and units);
3. scientific instruments and apparatus, including techniques of operation and aspects of safety;
4. scientific quantities and their determination;
5. scientific and technological applications with their social, economic and environmental implications.

Questions testing the above will often begin with one of the following words: define, state, describe, explain or outline.

B. HANDLING INFORMATION AND PROBLEM SOLVING

Students should be able, in words or using other written forms of presentation (i.e. symbolic, graphical and numerical), to

1. locate, select, organise and present information from a variety of sources.
2. translate information from one form to another.
3. manipulate numerical and other data.
4. use information to identify patterns, report trends and draw inferences.
5. present logical explanations for phenomena, patterns and relationships.
6. make predictions and hypotheses.
7. solve problems, including some of a quantitative nature.

Questions testing the above skills will often begin with one of the following words: predict, suggest, calculate or determine.

These skills cannot be precisely specified in the Curriculum Content because questions testing such skills are often based on information which is unfamiliar to the candidate. In answering such questions, candidates are required to use

principles and concepts that are within the syllabus and apply them in a logical, deductive manner to a novel situation.

SCHEME OF ASSESSMENT FOR CHEMISTRY FOR CLASS XI(CBSE)

Paper	Assessment Objective	Weightage %	Duration of Paper	Marks
Entrance to class 11	Factual Knowledge	10	30 Minutes	25 Marks
	Understanding of Concepts	30		
	Application Skills	40		
	Drawing Skills	20		

SYLLABUS OUTLINE OF CHEMISTRY FOR CLASS XI(CBSE)

TOPICS: MATTER- NATURE AND BEHAVIOUR; OUR ENVIRONMENT; NATURAL RESOURCES; SOME IMPORTANT CHEMICAL COMPOUNDS; PERIODICITY; CHEMICAL REACTIONS

MATTER - NATURE AND BEHAVIOUR

Nature of matter: Classification of matter based on chemical constitution- elements, compounds and mixtures. Homogeneous and heterogeneous solutions. True solution, suspension and colloids.

Structure of atom: Constituents of an atom, atomic number, atomic mass number, valency, isotopes and electronic configuration (upto atomic number 20)

OUR ENVIRONMENT

Environment and environmental problems: Causes, prevention and control of environmental problems (soil erosion, air pollution and accumulation of wastes). Biodegradable and Non-Biodegradable materials.

NATURAL RESOURCES

Metals: Metals, minerals and ores; General principles of metallurgical processes- Activity series of metals; General properties and corrosion of metals

Coal and petroleum: Coal and petroleum as natural resources of carbon and its compounds. Carbon- tetra valency catenation. Petroleum –fractional distillation. Hydrocarbon-homologous series, saturated and unsaturated hydrocarbons

SOME IMPORTANT CHEMICAL COMPOUNDS

Carbon compounds: Some synthetic polymers, soaps and detergents.

Acids, Bases and Salts: General properties of acids, bases and salts, pH as a measure of hydrogen ion concentrations.

Inorganic compounds: Preparation, properties and uses of washing soda, baking soda, plaster of paris and bleaching powder (outline only).

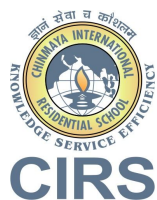
PERIODICITY

Periodic classification of elements: A brief historical perspective of periodic classification of elements, periodic law, modern periodic table of 18 columns, variation in properties across a period and along a group – metallic and non-metallic, atomic size.

Chemical Bonding: Formation of chemical bond, type of bonds – ionic and covalent bonds, electro negativity and polar covalent bond, properties of ionic and covalent compounds.

CHEMICAL REACTIONS

Chemical Reactions: Formulae of simple compounds, equations of simple chemical reactions and their balancing. Types of chemical reactions – combination, decomposition, displacement (single and double displacement), oxidation and reduction (in terms of gain/ loss of electrons).



CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL COIMBATORE - INDIA

ASSESSMENT OBJECTIVES FOR BIOLOGY FOR CLASS XI(CBSE)

The assessment objectives in Biology are:

- A. Knowledge with understanding**
- B. Handling information and solving problems**

A description of each assessment objective is as follows:

A. KNOWLEDGE WITH UNDERSTANDING

Students should be able to demonstrate knowledge and understanding in relation to

1. scientific phenomena, facts, laws, definitions, concepts and theories;
2. scientific vocabulary, terminology and conventions (including symbols, quantities and units);
3. scientific instruments and apparatus, including techniques of operation and aspects of safety.
4. scientific quantities and their determination;
5. scientific and technological applications with their social, economic and environmental implications.

Questions testing the above will often begin with one of the following words: define, state, describe, explain or outline.

B. HANDLING INFORMATION AND PROBLEM SOLVING

Students should be able to :

1. locate, select, organise and present information from a variety of sources.
2. translate information from one form to another.
3. manipulate numerical and other data.
4. use information to identify patterns, report trends and draw inferences.
5. analyse and explain a given data.
6. present logical explanations for phenomena, patterns and relationships.
7. make predictions and hypothesise.
8. solve problems using scientific principles.
9. use proper scientific conventions in tables, drawings and graphs.
10. pay attention to environmental impact.

Questions testing the above skills will often begin with one of the following words: predict, suggest, analyse, calculate, draw or determine.

These skills cannot be precisely specified in the Curriculum Content because questions testing such skills are often based on information which is unfamiliar to the candidate. In answering such questions, candidates are required to use principles and concepts that are within the syllabus and apply them in a logical, deductive manner to a novel situation.

SCHEME OF ASSESSMENT FOR BIOLOGY FOR CLASS XI(CBSE)

Paper	Assessment Objective	Weightage %	Duration of Paper	Marks
Entrance to class 11	Factual Knowledge	10	30 Minutes	25 Marks
	Understanding of Concepts	30		
	Application Skills	40		
	Drawing Skills	20		

SYLLABUS OUTLINE OF BIOLOGY FOR CLASS XI(CBSE)

TOPICS: NUTRITION, RESPIRATION, TRANSPORTATION, EXCRETION, CONTROL AND CO-ORDINATION, REPRODUCTION, HEREDITY, EVOLUTION, ENVIRONMENT AND ENVIRONMENTAL PROBLEMS & CELL STRUCTURE

NUTRITION

- Nutrition in plants- light and dark reactions of photosynthesis, factors affecting photosynthesis.
- Digestive system of human beings.

RESPIRATION

- Breathing and respiration.
- Aerobic and Anaerobic respiration.

- Respiration in plants.
- Respiratory system human beings.

TRANSPORTATION

- Xylem and phloem – components and function.
- Composition and functions of blood.
- Circulatory system in human beings.

EXCRETION

- Excretory system in human beings.

CONTROL & COORDINATION

- Co-ordination in plants.
- Human nervous system.
- Endocrine glands, their hormone and functions in human beings.

REPRODUCTION

- Types of asexual reproduction.
- Pollination and fertilization in flowering plants.
- Male and female reproductive system in human beings.

HERIDITY

- Chromosomes, genes and DNA.

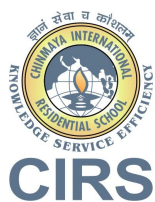
EVOLUTION

- Evidences of evolution.
- Theories of evolution.

ENVIRONMENT & ENVIRONMENTAL PROBLEMS

- Air, water and soil pollution and their control.
- Sustainable development.

CELL STRUCTURE AND FUNCTIONS



CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL COIMBATORE - INDIA

ASSESSMENT OBJECTIVES FOR MANAGEMENT STUDIES FOR CLASS XI (CBSE)

The entrance tests is designed to objectively measure the aptitude of the candidate, and the general skills as expected of a would be manager / organizer.

The candidate is expected to:

- have general awareness of the current happenings in the Economy as well as in businesses.
- Be able to interpret basic charts, graphs or tables in relation to business activities and / or the Economy.
- Be able to comprehend information given in case studies and suggest logical solutions to it.
- Be able to do simple calculations on profit and loss.

SCHEME OF ASSESSMENT FOR MANAGEMENT STUDIES FOR CLASS XI (CBSE)

Assessment Objectives	Weightage (%)	Duration of Paper	Marks
Factual knowledge	30	1 Hour	50 Marks
Understanding of concepts	30		
Application skills	20		
Problem Solving skills	20		

SYLLABUS OUTLINE OF MANAGEMENT STUDIES FOR CLASS XI (CBSE)

The candidate is expected to have a **general awareness** of the following topics:

Marketing

- How and why businesses advertise their products?

Human Resources / Personnel Management

- Ways to promote positive work climate

Profit and Loss

- Simple calculations
- Basic Data Interpretation of charts, graphs or tables

Economy

- Basic features of Socialist, Capitalist and Mixed Economies
- Basic features of Developed, Developing and Underdeveloped economies

Challenges faced by an economy and its effects

- Population
- Child labour
- Poverty
- Basic Data Interpretation of Charts, graphs or tables